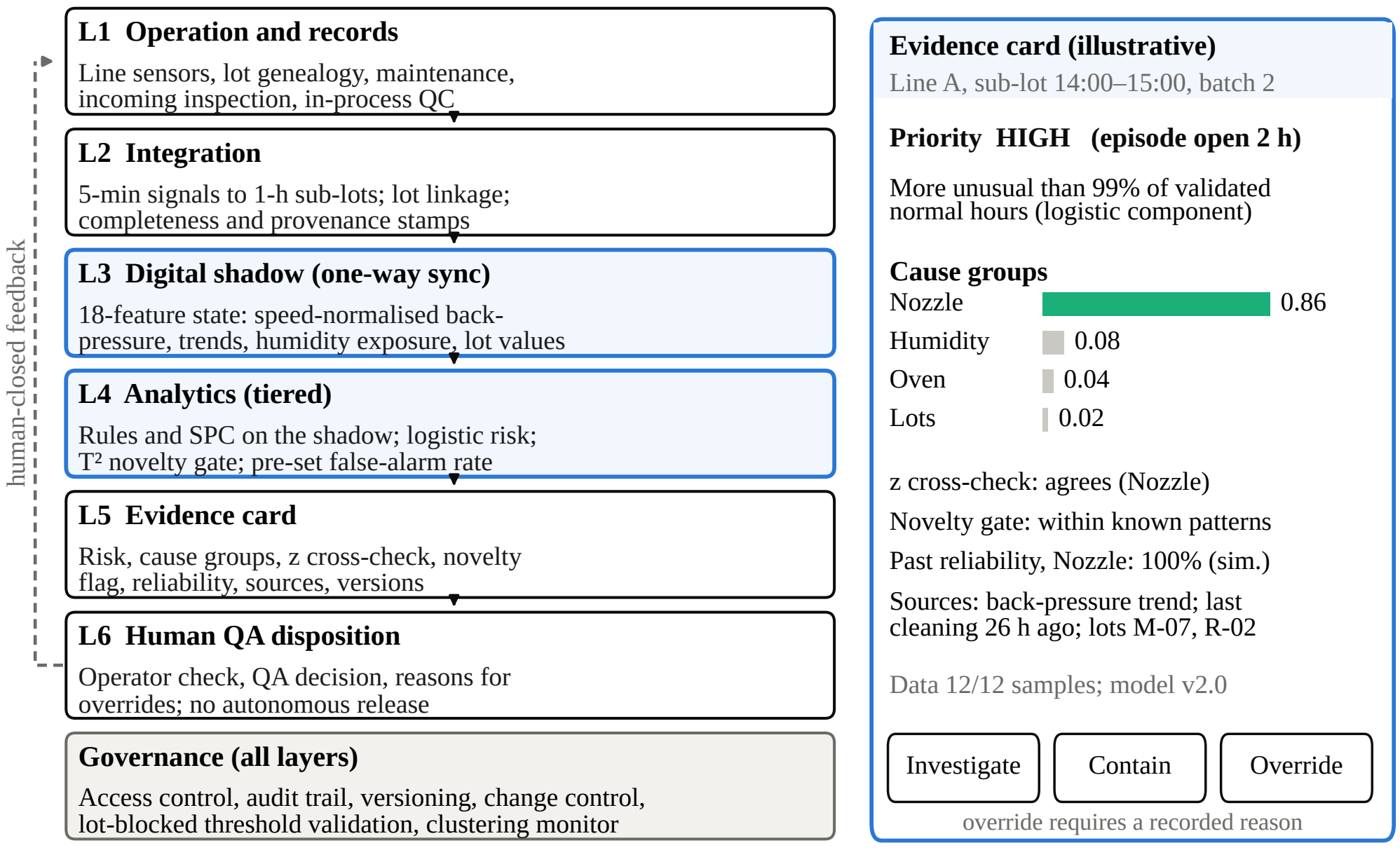
L1 Operation and records
Line sensors, lot genealogy, maintenance, incoming inspection, in-process QC
L2 Integration
5-min signals to 1-h sub-lots; lot linkage; completeness and provenance stamps
L3 Digital shadow (one-way sync)
18-feature state: speed-normalised back-pressure, trends, humidity exposure, lot values
L4 Analytics (tiered)
Rules and SPC on the shadow; logistic risk; T² novelty gate; pre-set false-alarm rate
L5 Evidence card
Risk, cause groups, z cross-check, novelty flag, reliability, sources, versions
L6 Human QA disposition
Operator check, QA decision, reasons for overrides; no autonomous release
human-closed feedback
Governance (all layers)
Access control, audit trail, versioning, change control, lot-blocked threshold validation, clustering monitor
Evidence card (illustrative)
Line A, sub-lot 14:00–15:00, batch 2
Priority HIGH (episode open 2 h)
More unusual than 99% of validated normal hours (logistic component)
Cause groups
Nozzle 0.86
Humidity 0.08
Oven 0.04
Lots 0.02
z cross-check: agrees (Nozzle)
Novelty gate: within known patterns
Past reliability, Nozzle: 100% (sim.)
Sources: back-pressure trend; last cleaning 26 h ago; lots M-07, R-02
Data 12/12 samples; model v2.0
Investigate
Contain
Override
override requires a recorded reason